DIFFERENTIAL EQUATIONS

BLAKE FARMAN

Lafayette College

Name: Solutions

Find the solution of the differential equation that satisfies the given initial condition.

1. $y' = xe^y$, $y(0) = 0$.

$$\int e^{-y}dy = \int x\,dx$$

$$\Rightarrow -e^{-y} = \frac{1}{2}x^2 + C$$

$$\Rightarrow -y = \ln\left(-\frac{1}{2}x^2 + C\right)$$

$$\Rightarrow y = -\ln\left(-\frac{1}{2}x^2 + C\right)$$

$$0 = -\ln\left(-\frac{1}{2}0^2 + C\right) = -\ln(C)$$

$$\Rightarrow C = 1$$

$$\boxed{y = -\ln\left(1 - \frac{1}{2}x^2\right)}$$

2. $y' = \dfrac{x\sin(x)}{y}$, $y(0) = -1$.

$$\int y\,dy = \int x\sin(x)dx$$

$$\Rightarrow \frac{1}{2}y^2 = -x\cos(x) + \sin(x) + C$$

$$\Rightarrow y = \pm 2\sqrt{\sin(x) - x\cos(x) + C}$$

$$-1 = \pm 2\sqrt{0 - 0 + C} = \pm 2\sqrt{C}$$

$$\sqrt{C} > 0 \Rightarrow -1 = -2\sqrt{C} \Rightarrow \sqrt{C} = \frac{1}{2},\ C = \frac{1}{4}$$

Therefore

$$\boxed{y = -2\sqrt{\sin(x) - x\cos(x) + \frac{1}{4}}}$$

Date: March 21, 2019.

3. $P' = \sqrt{Pt}$, $P(1) = 2$.

$$\int P^{-1/2}\,dP = \int t^{1/2}\,dt$$

$$\Rightarrow \quad 2P^{1/2} = 2\sqrt{P} = \frac{2}{3}t^{3/2} + C$$

$$\Rightarrow \quad \sqrt{P} = \frac{1}{3}t^{3/2} + C$$

$$\Rightarrow \quad P = \left(\frac{1}{3}t^{3/2} + C\right)^2$$

$$2 = \left(\frac{1}{3} + C\right)^2$$

$$\Rightarrow \quad C = -\frac{1}{3} + \sqrt{2}$$

$$\boxed{P = \left(\frac{1}{3}t^{3/2} + \sqrt{2} - \frac{1}{3}\right)^2.}$$

4. $y' = -x/y$, $y(0) = 3$.

$$\int y\,dy = -\int x\,dx$$

$$\Rightarrow \quad \frac{1}{2}y^2 = -\frac{1}{2}x^2 + C$$

$$\Rightarrow \quad y^2 = -x^2 + C$$

$$\Rightarrow \quad y^2 + x^2 = C$$

$$9 + 0^2 = C$$

$$\Rightarrow \quad C = 9$$

$$\boxed{x^2 + y^2 = 9}$$